2 – August Focus Group – FG3

Interviewer

So, to really kick things off and try and work as a bit of an ice breaker just to introduce everyone, we'll just kind of go around and just say what diabetes technology you're using, what smartwatch you're using as part of the study as well. And then one bit of type one diabetes technology you dislike or have disliked in the past? Um, so I'll go first. Um. I mean I've had type one diabetes now for 20 years or so. Um. I recently moved on to the Tandem t:slim X2 and the Dexcom sensors. I'm having to self-fund the G6 sensors at the moment, which is a bit of a pain, but I'm hoping the NICE guidelines come out soon and kind of sorts that out. So hopefully that will take that away. Um, I've actually been using a Fitbit Sense. I got on this kind of the start of the PhD to get a feel for how they work and kind of get that perspective moving in from it. And then my bit of type one diabetes technology I'm not a fan of I have here. Is well, it's the Aviva Combo. So, it's an old blood glucose monitor. So, it used to have the Spirit Combo pump and they were Bluetoothed together, which was really good. But you realised that every time you use it tries to connect up to the pump, which normally takes a while. And so I'm still using it now, despite not using the pump and every time I turn it on, it takes about 15 seconds to connect to a pump that's no longer there, which is a massive pain and slows everything down. And I'm glad I'm not having to do it that often. Uh, but yeah, it's just one of those pains I guess you have to deal with. Um, so yeah, yeah, we'll kind of go around the room now. So. Yeah. [P04], do you mind going next and introducing yourself?

P04

Um, hi I'm [P04], I've been type 1 for five years I think now. Um. I'm on t:slim and Dexcom G6, which I've luckily got both on the NHS. I'm not quite sure how I managed that. Which I find great. One thing I don't like was really didn't like Libre 2. I found it was really, really inaccurate for my blood sugars and would mean that I would correct when my bloods would definitely not low and correct when my bloods are definitely not high, which I found ended up being a complete pain. So yeah.

Interviewer

Yeah, yeah, that's definitely sounds like a pain. Do you mind picking someone to go next?

P04

Uh, [P17]?

P17

I'm [P17] and I've recently just gone on to the Omnipod 5. Um. So, I'm doing the hybrid looping with the Dexcom G6, which is really good. And I've had diabetes 16 years next month and a piece of kit that I really don't like. It's kind of two is kind of the Libre 2 agree with everything you've just said.

And I was on the Guardian sensor as well and I still hear that beep anyway, I still kind of react and panic because it would just beep all the time. So, I'm glad to get rid of it.

Interviewer

Yeah. No, it's super interesting to hear you're on the Omnipod 5. Yeah. I'm, I'm interested to hear how it's been going. Um yeah, which leaves [P11]. Do you mind going now?

P11

Yeah. I'm. Hi, I'm [P11]. I've also been type one for about 5 years and I'm currently on the Medtronic something, something, but I'm about to go into the Tandem like in two week's time. And so, I've also got the Dexcom, so I'm excited to try the like IQ technology and hopefully that will help with some of my like high blood sugar. And yeah, I was also on the Libre and hated it. But I think also because for some I still have to use it because for some reason my like blood glucose strips, they changed them all to the freestyle ones, so now if I want to do a blood test or ketone test I have to use the Libre. So, I feel like I'm carrying around like double technology at the moment cause they're still using the Libre device but with the Dexcom anyway. Yeah, that's me.

Interviewer

Oh brilliant. I feel sorry for Abbott, they're taking you a bit of a beating this call. Um. Yeah. So I guess just a little bit about how this will work. I'll just kind of throw questions out there, but it would be really great if someone says something other people kind of jump in and say if they have the same, if it's different un and really build from there. I I hate when people do this, but I'm really interested to hear what everyone says, so maybe being a bit quieter might come and pick a you to kind of say how you feel about stuff. If there's nothing to say, just say, it's absolutely fine. We're not here to like force you to say anything. It just be interesting to hear as much as we can about how you've been finding it. So yeah, kind of um to kick things off then, it just kind of think about the smartwatch over this second month you've been using it or potentially longer, I guess people are using different ones. Yeah. Just what feature do you think you're kind of using most currently or have been using most currently this month?

P04

I think the one I've been using most recently and I think on my Fitbit app, it said that I had access to six months free Fitbit Premium. So, I was like, I'll give that a go and it does loads of stuff about sleep. And I find that my sleep. I always feel tired and I seem to get up a lot in the night, so I thought I'd start wearing it overnight and see how, see if there's anything that affecting my sleep, and it does, like restlessness score and things like that. So, I've only been doing it for a few days, but so far it's been quite interesting to see whether like certain things seemed to impact how I'm sleeping and things like that. So I've been finding that quite useful to use.

Interviewer

Yeah. Does anyone else use the sleep side?

P11

Yeah, I was going to say I agree with that. I've used it a fair amount or just like logged. I haven't got the premium. I think I could get the premium, but I haven't like accessed it yet if that makes sense. Yeah, but I find it interesting cause I usually get up to go to the loo at least once a night and it seems to be like at exactly the same time each night, so I'm trying to work out what it is. It also tells me, obviously when my blood sugar is high, so, I'm like, what do I need to change my basal rate and stuff? But it's just quite interesting to try and see where that is, but also I've been on holiday and I've been a lot more active than usual. So, I've been like using the exercise things and trying to see if it makes a difference cause it does all the different like how fast you're going like the different zones and stuff, and I'm trying to, I still haven't worked out, but like if I'm in a different zone, is that affecting things a different way anyway? And so yeah, kind of be looking that bit more.

P17

I think just linking to what you said with the sleep, so I was on the Fitbit, but it was irritating my skin so I've gone back to my Apple Watch. Um. And one of the things that I'm kind of gutted about was the, I know I've said to [Interviewer], that the battery life on the Apple Watch is awful compared to the Fitbit. But since I've been on the Omnipod, my sleep has been so much better just cause my sugars are much more controlled through the night. So, it was the exact same getting up. But even with my Apple watches kind of just picking up when I'm taking it off and when I'm putting it on, first thing it's coming up with trends saying I am sleeping for longer. But it's not analysing like the different stages of sleep. Um, so I'd definitely see that. And then kind of activity and looking at the trends. So, it keeps notifying me with different things and it were about three days ago and I've never seen this on there before, it notified me that since I've been on the Omnipod it's picking up my Dexcom readings. So, it was telling me, "during your last workout, you're sugar stayed between like X and Z" kind of thing, which is quite interesting to see. So, keep logging on to see what the difference is.

Interviewer

Yeah, that sounds really interesting. Yeah. Do you mind talking a bit more about that. What is that through the Apple Watch itself?

P17

Yes. So I have notifications with the health app and it kind of just popped up on my phone and it was like, look at your your glucose levels. And I I think what I had Dexcom I kind of say yes to everything to link it to all the apps. But I didn't realise it really, it was taking it into like the health app. So it comes through and I'll see what I can see what my last one said now. Um. But it was coming up with it different things of like when you were walking. So every time my Dexcom reads and connects to my phone it all the data was going through that, the health app will tell you like how many times you've inputted things. Um. So if I go and let the last week. So, like it's telling me now for my workouts my

sugars have been between 4.3 and 9.4. But, if you've worked out that date it'll kind of tell you as well, so it's picking up on trends. Um, so don't know what it will do with the trend, it'll be interesting to see the more I wear it. But it's quite nice to see that it's interlinked a lot more compared to the other sensors I've been on.

Interviewer

Yeah, definitely. I think a few people commenting on, yeah, kind of seeing this information, like how much do you actually change things as a result of it? Or have you made any adjustments because of what you've seen?

P17

I think because I'm still playing around with settings with Omnipods. I'm having weekly check ins at the moment. There's like an activity mode as if you were gonna put like a temp basal on, if you were on a Medtronic pump. So, I've kind of been trying to play around with that and and like the algorithms and things. But at the moment is kind of just been steady. It's just making sure that my sugar is a little bit higher before I go, but it's been quite nice to see that it's just recording between the time I start the activity on my watch and see what it records.

Interviewer

Yeah, what about everyone else? I guess that was quite mention of the sleep side of things as anyone just things based on what they've seen?

P11

Ohh not really so far. I think it's. I don't know with sleep it's kind of harder to, cause I don't feel like I'm in control of how well or not I sleep. I kind of just feel like it just happened. But, definitely with the seeing the readings on the watch I found that. So, like when I was on holiday and out like all day on my feet all day, that was super helpful because then I did put temporary basals on and stuff as a result of seeing. Like if I could see I was going down and I knew I was going to be walking for like another two hours, so I was like "OK", you know. So, I think it's something that's just having the visibility there really helped without having to get my phone out and all that kind of stuff. Cause especially when I was I was in cities and stuff and you know, you're always told don't get your phone out because of pickpockets and all that kind of stuff. And actually, I just didn't have to, I had it in my bag and like that it just kind of took an extra worry thing away because then I could react without having to constantly be aware of where everything was. So yeah, that was good.

P04

Yeah, I think like the activity side of it, I've got mine set up so I can just like flick my wrist and my blood sugars are they're on the front screen. And kind of like I've always done, like a run as a warm up and then done like the weight stuff afterwards. But actually, I found now that that seems to make

my bloods rise a bit cause I have the run and it drops my bloods and so I'm like “ohh, before I do something else, I better have something to eat quick”. And then suddenly before I know it, I'm ending up high. So, kind of being able to see that I'm actually going, “ohh, maybe I should swap this around and try a different way. Maybe try and put my activity mode on sooner than I usually would” and things like that. Yeah, I think that's been really helpful.

Interviewer

Yeah, that's good to hear. Is that, so everyone's got the readings coming through on there wrist of some kind? Um. Yeah. No, I I guess there are. What what are the times you find that useful? I guess there is kind of the walking out on holidays mentioned and in the gym and there any of the times you find that particularly beneficial?

P17

Um, I would say for me, if I'm in the office um, because I've got the Omnipod now and I've got it, I usually got my phone on me for like to check my sugars and now I've got the reader as well for the Omnipod. I felt like I'm carrying a lot round with just having my watch and kind of just glance down and see without. Because as soon as I mentioned you were diabetes in the office, everyone goes into panic mode and thinks that they need to intervene and help them, whereas I just want to see where they are, it's just fine.

P11

Yeah, definitely. I found at work as well. So, I'm a teacher, so I'm like dealing with kids all the time and I just don't have my phone on me cause like we're kind of not allowed to have a phones on us. But I can just have it in my bag and then like, in between lessons, if I've only got 5 minutes. Like, you know, it's just so much easier then. It sounds stupid because it's on my phone home screen as well but it's still a barrier with, like picking up your phone. Whereas, if something's just literally on your watch and it's not a taboo thing because people without diabetes check their watch as well, like, it's just, I feel like I'm not drawing attention to it. And again, at work, no one's no one really knows. And what I mentioned it to them and they were like, “ohh, we had no idea you could do that”. And so it's just not an obvious thing. So it doesn't draw attention to it as much, which I really like.

P04

Yeah, I was going to say at work as well. Um I'm I'm just working over summer before I go back to uni, but it's really easy just to be able to be like I'm doing something, I'm walking around like ohh yeah I'm still fine and things like that. And I think then when I go back to uni kind of in lectures and on placements and stuff, being able to just like flick my wrist rather than having to be like “ohh sorry, give me two moments to go and find my phone” and all that sort of stuff, I think it has been so much easier.

P17

Another one as well, I was just going to say is driving. Just being able to, cause he's on my watch, just kind of just turn my wrist and it’s there, rather than pulling over and trying to scan like when I use to scan my arm or open my phone. And with the Dexcom as well, I've added it on Siri, you can ask Siri as well. So either or in the car is really helpful.

Interviewer

Yeah, it seems people are talking about like the ease of doing it. But I guess those kind of another level which is kind of almost the discretion, so it's not so obvious to other people, is that something you'd prioritise and your diabetes technology?

P11

For me, I think yes, especially having gone from the Libre where like it just.

Interviewer

Ohh you muted.

P11

Ohh sorry, I was just gonna say with the Libre, it's so awkward having to do like this [gestures scanning their arm] like and everyone's like, “what are you doing?”, because no one uses their phone like that. But with the with the Dexcom on the watch, like no one knows. And I think I think maybe especially sort of young people and like you kind of fitting in and you're doing new jobs and like. I've just finished uni as well and like you know I was out clubbing and like you don't wanna be like a burden to anyone. And there's quite a lot of new social situations that I find myself in and I just don't have to explain it anymore, which is really nice.

P04

Yeah, I think.

P17

I think the problems ohh sorry.

P04

No you go. No you go.

P17

And I was just gonna say with the alarms I know, especially like the Libre 2, that I would find they never even worked. Whereas the Dexcom, if my phone's on vibrate, my watch will vibrate rather than this big alarm going off and everyone's left panicking around me. Even if it's people I know, like my partner, he would wake up all the time because like my other sensors and things would be going off playing this high pitch alarm. I'm obviously used to them and I used to sleep through them so they weren't helping me at all, just keeping everyone in my house up. But with the Dexcom, it's so subtle, like if I do like where my watch to sleep or even my phone it comes up and rather let all these alarms going off. Um, obviously they're still there but as and when you want them.

P04

Yeah, I think kind of like. I think people around me seem to be pretty switched on about it anyway, but like, if I'm at work I like how it's not, I can just see it, that it's gone low. I don't have to deal with the massive alarm going off and scaring everyone like you said. Um. And kind of just like usually they've clocked by the time I've got out like a juice carton or sweets or whatever cause it's the only time I ever eat and drink those. But it's kind of like it's less like "oh my God are you okay" sort of thing it's just kind of like "ohh yeah you're fine you're sorting it yourself", rather than this like blaring along going off so.

Interviewer

Yeah, I kind of like speaking about the alarms. I I definitely relate to that I've been kicked onto the sofa a few times because of those in the middle of the night. Um, but yeah, how do you find. Is it different coming through the watch or do people use the alarms on the watch versus the phone before?

P11

For me, for somehow on the watch it's vibrate. I'm not sure what setting I've. Yeah, I probably need to look at the settings, but on the watch, it vibrates even if my phone is on loud. But I quite like that again when I'm asleep. It probably contributes to me waking up cause my blood sugars aren't very stable at the moment, so it does mean I wake up, but it also means I wake up. Like there's a negative to it cause it means I don't ignore it because it's vibrating on my watch, it's much harder to ignore than the the alarm. But, it does mean that my sleep is probably more broken. I have found like if I'm sitting at I don't know 11 or something and then I suddenly go up to 12 and then come back down then maybe just go. So, it's actually steady, then it gets quite annoying because it's like it'll buzz every time and that probably doesn't help with kind of the sleep aspect. But on the whole, it doesn't make a noise and again, I quite like that cause I feel it, but no one else around me does.

P04

Yeah, I find that even though I've been listening to alarms for five years now, like they still make me jump a lot of the time, cause they are really loud. Um. Snd so actually having it just buzz, it doesn't make me jump, it doesn't make people around me jump and panic and things like that, and I just think it's it's just so much easier just being able to be like, "ohh I buzzed. I know that's that". Like I've

got mine set up so that's the only thing that buzzes on my wrist. So, even if I'm not looking at it, if it buzzes, I know it's to do with my blood sugars and then I can just go and sort it out then um rather than having to like check it all the time. So, it's quite nice.

P17

I think yeah, with the alarms. Um. So I was in like a fitness class the other day, but I was like helping to instruct it and my alarm on my. So on my Omnipod, the only alarm that will go off on that device is if you're low. So, my watch was like set it up, so it was vibrating when I knew it was it's what it's like right, I looked and like 3.4. So, I can go and deal with it. But as I was going to pick my phone was beeping at me, my Omnipod controller was beeping at me and my Omnipod. And like everyone, I was kind of teaching was just like panicking. All these alarms were going off and like, it's fine as soon as I pick the orange juice up, like you said, everyone knows you're OK. But it was kind of a bit of a reminder cause I have been it's been so new not having like alarms. Because my sugars used to spike up and down all the time and because they are a lot more steadier and in range I'm not experienced as many, but it's kind of brings you back. And when I was on holiday as well, could you other people's alarms going off, and I was like, “I know that sound” and there was a Medtronic pump on someone like sitting on two loungers up going off. Even though I wasn't even on the pump at that point.

Interviewer

Yeah, yeah, yeah. I guess that's kind of that's the other side to it as well is it is another device where you can have information or alarms coming through. Yeah. How do you feel about, yeah, there being that extra source and those extra details you can see if you want to?

P11

I don't know, for me, I think I spoke about this last time, but there have been days where I've chosen not to wear it because I don't want to see all the information. Um, but that's that's a me issue. Like there are days where it just trying to avoid it. And and so I don't and I don't find some of the other stuff on the Fitbit particularly helpful things like tracking what you're eating and you're weight and all that kind of stuff. Like for me those kind of reminders, which I think somehow I turned all the reminders on and like for for a while everyday it was asking me to input my weight and stuff when I was like “ohh no, I really don't want to have all that data”, like that's not healthy for me. And again, with the diabetes data some days like it's too much and I I know it's on my phone I'll check it when I need to. And then there are other days where especially if I'm really active, I want to be able to check it all the time. But for example, today I know I'm like literally in my house doing chores, working. You know, I haven't got it on today because I'm like actually today I need a chill day and I I can check my phone. I know my phone would alarms will go off. So I don't know, I think I've been kind of using it as an extra, rather than a replacement, I thin. And just I'm trying to be really cautious about what I turn on and what I don't cause I think the information can be a bit too much for me, but that's probably personal to me.

P04

Yeah, I think I've I've think I've definitely started using mine as a replacement. Like there's been a couple of days where I've forgotten to put it my watch on and then I spend my whole day trying to check my bloods on my wrist, and it's obviously not there. So, I think, yeah, I think I've definitely started to use. I find it very, very helpful just having it right there. Because where I'm on t:slim now and it's all closed-loop, it's definitely much more stable than it was before I started. I've only been on t:slim for a couple of months now and so I kind of like don't need to necessary. I'm not one to kind of like, sit on near my phone a lot anyway. So actually, being able to just like, flick my wrist, I find it really useful or it just buzzing and I'm not don't have to think about it for the rest of the day. I much prefer that than having to constantly kind of like think about "ohh where's my phone" and let me double check that they're right and things like that. So, I've definitely replaced a lot of the time I'd spend on my phone. I use my watch, which I quite like.

P17

I think I'm a bit of the opposite to what you said, [P11], where I kind of, I was saying to [Interviewer] in the last interview, having the watch and since that the Dexcom being on there, I'm constantly looking all the time and I think it has become a bit replacement. So, when I first went on it and I think especially cause I kind of started with the Dexcom like a week before my pump, there was a lot of change in technology for me. It was like all kind of new and I think I was quite inquisitive to everything but it was constantly checking all the time to see where my sugars are. I've got a bit more relaxed now I'm like a month in, but I think if I haven't got my watch on now, I'm kind of again trying to look to see like on my wrist and is not there. So, I'm getting quite reliant on it. Um, so I think, yeah, if my watch broke, I think it would be a bit lost.

Interviewer

Yeah. How do people feel about that reliance? Is that, do you see it as a good or bad thing or?

P17

I think it's become quite comfort for me. And I think I'm trying to think like when I was younger when my HbA1c was like 100 plus growing up, when I was quite against my diabetes was very accepting it. I think if I had something like this really was more subtle and annoying like the school canteen getting a full glucose kit out and asking for it to be excused. If I could be looking at my wrist and seeing, oh I need to make an, take an injection, I think my control could have been a lot better. So, I think, yeah, if I'd had it for like my younger that it definitely would have helped.

P04

Yeah, I find it's kind of like it just means that I'm not having, it can kind of be at the back of my head rather than like all I think about all day. Kind of being like "ohh I need to double check if my bloods are good". Especially like with things like finger pricking, I've had a couple of days where I've been at work and I've accidentally ripped off my Dexcom um cause it's an old one and it like it needs changing a couple days. And sending ripped it off and then like suddenly having to go back to fingerpricking and stuff. I'm definitely like very reliant on the facts and now that I can kind of just like

quickly check it and not have to think about it. And like even that two-hour period where it warms up, I spent my whole time kind of finding myself like checking my wrist and being like “ohh yeah, no, I can't see it”. Or even on my phone like swiping and being, “ohh no, I can't see it”. So definitely reliant on the fact that I’m able to check it quickly. I do think it means that I look at them more often but I don't think that necessarily means that it’s a bad thing. I think that just probably means that I will be slight. I'm not too reactive with it anymore, anyway, cause I've got the pump but like if I wasn't on a pump I think it would probably cause me to seesaw a bit cause I kind of be a bit more reactive to, “oh, my blood's are going higher, better give some insulin” or “my blood starting to go low, better have some sugar”. Whereas now, I kind of like try and let the pump sort it out itself so. I think if I had it and didn't have a closed-loop, I'd probably see it as much more of a bad thing, whereas now I kinda just like. It's just a reminder rather than kind of something I'm using.

P11

I think, yeah, I think that's maybe why I'm slightly different because I'm not on a closed-loop system yet. So, I think for me, my blood's do you go like up and down. And so there are days where it could just constantly be buzzing and at that point, I'm just like, I need it, I need it off. Or because it's also like if I've sorted something out, like corrected a high, I know it's gonna take two hours for that to kick it in. Do I want it to buzz every 5 minutes for two hours? No. So especially like if I'm asleep, or if I'm, you know, I don't know. I think I'm still trying to work out but I'm hoping cause I'm going on the tandem in two weeks time it will get better. But I think. Yeah, it does kind of scare me though how reliant I become on technology. And again, I don't think it's a bad thing, but it does worry me that then if something went wrong I would, I get super anxious if I don't have my phone or my watch, which is the same I think it's what you're saying. If you can't, if you then can't see it, especially when it's warming up, the anxiety like spikes through the roof, so I don't know. I think like the majority of the time it's fine, but if for some reason then it disconnects or I'm immediately on guard. So yeah, I don't know, I think that yeah, pros and cons, but mainly pros.

P17

I think just thinking to what you just said then about like kind of anxiety going on them. That was literally me for like the first week when I went on Omnipod. So, kind of watching it like reading your sugars and the algorithm as it was building. Whereas my sugars would always be spiking, I was finding, when I went on my Omnipod I was low for like the full session. And even though I was in a room, I was just trying to discreetly eat glucotabs, even though I'm in a room full of diabetics. But when I left and I had food and I was kind of putting my correction in. Um. Normally if I went to McDonald's you always, I never give like the full amount of carbs because I know like that my sugars never react straight away. And they were like “just go back to complete like new carb counting”. And remember it was like 95 grammes of carbs and it was like 20 units of insulin going in and I was like, “I know I'm just gonna plummet”. And as it went in, I could just see it like going up and I was like, “I can't give it correction because it's gonna do it for me”. And remember just staring up my Omnipod reader for like half hour waiting for it to do something. But it took so long to get kind of put the trust in to it. But I do worry if, like a couple of times by pods have fallen off and then I know there's no insulin, it thinks there's insulin my system, but there's not. But it goes back into manual mode as soon as you back into it. You go always have to go back into your old ways of working on a pump. So, it's kind of just keeping it in the loop that is still there. But like you said, if you haven't done it for a

while, it's kind of just remembering to go back into that. So, that's been a big learning curve for me this month.

Interviewer

Yeah, that's quite interesting. Yeah. You saying about this kind of process of onboarding on to closed-loops and I guess, yeah, so you've go through it recently and [P04] previously and [P11], you're just about to. Is there anything? Ohh, what bits that process made it easier for you or what bits would you like to kind of would you like to have seen more of or would have helped?

P17

I think for me when I went on it, they were like, "come up to the hospital, like we'll set you up on it", and I thought it was gonna be kind of in and out and going on a a wireless pump and it was gonna be quite straight forward. And there was like, a two-hour presentation of all this information. It was like when I went on the DAFNE course. And I think you take all this information, and they're like, "right, okay off you go". And it was kind of digesting and I was like, "right okay I understand it all, but I'm not gonna remember when it comes to a point where I need to apply that". So, it was kind of they give you information and that that the help from like Dexcom and Omnipod on the e-mail and things have been amazing, I get a reply within an hour. But it was kind of if something goes wrong during the night now and you're kind of just trusted to go and work with all these things. So I think maybe kind of introduction sessions rather than being like use everything off you go. Because I know there was a girl next to me there was up there was a first time of her being on a pump, so she even just words like bolus, basal and kind of getting used to things like that. I just think I don't think I could have gone straight onto like a full like loop system with that kind of knowledge. So, just breaking it down maybe to allow you to digest everything before going on it.

P04

Yeah, I think with my pump, I'm very, very worried about it failing in some way. It's like I used to be on Omnipod, obviously not Omnipod 5, just the normal one. Um. And I went my time on it I went through three handsets and cause it would just suddenly stop working. And then obviously you phoned them and they like get it sent out to you like that, like within an hour and you get it within a couple. But it's such a stressful moment cause you, because all your basal insulin is in Novorapid or whatever, I think mine is. And it's like suddenly I'm like "ohh, what do I do if my blood start going high? I've got no background insulin going in anymore". And obviously with t:slim you just get your handset that's connected to you and I'm like if that goes wrong or it breaks or whatever, like I'm worried about how quick I'll be able to get a new one and what I do because of that. Because I don't think they really teach you like how, what to do if it stops working. Like I remember when I was a part of the paediatric team, they were kind of like, gave me a set number for my basal and said "ohh well like you can just go back to this if you ever need to". But it's like actually now I'm older, does that mean that that basal won't doesn't match what I do now? And because it's a completely different type of insulin, is that not gonna work if I try and put the same amount of basal I technically have now? And it's yeah, I think. I much prefer the pump I'm on, but it's kind of like it seems like a bit more of a "ohh no", if it goes wrong, but we'll see.

P11

Yeah, I think it's the same for me. Like the worry, the technology is getting better and better and like I know that when it go on the t:slim, it will do amazing things to me because it will auto correct and all that kind of stuff. But I am worried about becoming complacent with it. And like, obviously I'm looking forward to just being able to look down and not have to do anything and be really reassured. But then it's like, as you say, if something suddenly goes wrong, you're not using the skills that you're building up, that I'm currently having to use everyday and like the mental brain power of knowing how much to correct and all that kind of stuff and I am worried about that. I think especially having just come back from travel and like, I don't know whether you guys do this, but having to go through all the airport scanners. And I always I was always taught like you can't go through them, which I think has now changed but anyway having to explain that in like different languages and being challenged on that and like I found it really really stressful. And then I got really anxious and they were like "why are you getting anxious?" And like because if if this breaks now, I'm in a different country, like I don't, you know. So I think, yeah, I think that's the the background. I feel like day-to-day it doesn't impact me, but then you suddenly get into situations and then suddenly it's like going from really low anxiety to super high anxiety rather than having like a daily middle level. So yeah.

P17

I think what you.

P04

Yeah, I I I feel you. I went last year, while I was still on my Omnipod and stuff. I was Omnipod and Dexcom um coming back through Greek airport security and trying to explain to them that you can't go through the scanners cause it will break the stuff. And my bag can't go through the X-ray and I know I understand it's got loads of needles in it and that's really weird, but please don't put it through the scanner it's gonna break. And I literally ended up bursting into tears in the middle of airport security, so I understand how you feel.

P17

Think what you said about the you know the pat down and different things. When I did my Omnipod training, they told me that you can go through any of the scanners now apart from like the certain like X-ray ones in like in the hospital. So, all the ones I've always gone into airports and just said because I can never remember between the different sensors and and different things I've been on I'd rather be patted down. And I've only ever had one experience where I think I was coming back from [UK city] and they were like, I was on the Medtronic, and they were like "can you just take it off, it's easier". I'm just like, "no, I'm not just gonna take it off". But I think I'm going, I'm going away on Monday, so it'll be interesting to see what it's like travelling with the the Omnipod. But I think even now that I know everything can go through, the scanner is just for my peace of mind you know when I get there. Because I think if something goes wrong when I'm on holiday I'm immediately gonna think the airport has messed up everything. So, I'll just keep my kind of stress levels down just for my peace of mind.

Interviewer

Yeah, definitely. I think it's. I mean, it's still these events that come through where things broke down. I wonder with the recent, some of you may have heard about this, but then the Libre app recently breaking on iPhone's when they tried to do the update and it’s not comes through and I know some people were left without kind of a functional app to use. I just wonder if what your reactions to that were and what you think can be done to kind of mitigate from those scenarios.

P04

I think.

P11

I think I was just glad. Sorry you go.

P04

Yeah. I was just saying I was glad I wasn't on Libre anymore.

P17

I went on Dexcom two weeks before that happened. I went on Dexcom literally just before I went on holiday and the day it all crashed was the day I was gonna be coming back through the airport where like connecting flight. And like I just remember kind of go on Instagram as I got on and off the flight and seeing it all. And I was like if that would have been me on a flight now with no way, because I know it would not have known where my finger pricker was to kind of find, I know would have been in panic mode. So, stuff does like that does worry me if it happens on anything else I'm using.

P11

I think it's. It's weird, isn't it? Cause I think you're always. I don't know whether you were, but when I was first diagnosed, like they taught me on, like, injections and fingerprick. So, like, I know there's a backup system and I always take the stuff, but I don't trust it as much as I do the technology now. And so. Yeah, it's just weird cause I know I know I could do it, but I don't feel like I've got the skills I almost feel like I need a renewal class on the old system as I go on to more and more technology. Because I think actually, yes, that'd happen with the Dexcom, yes, I can finger prick, but I know now that that is giving me one moment in time and like 5 minutes later I could be completely different. So having been on the Dexcom, you know how unreliable the fingerprick is. So it's. Yeah, it's scares me. Yeah. If something were to go wrong.

P04

Yeah, I think that's the thing. It's like you get so used to on the Dexcom being like, OK, so my blood's are 6, that's great, but actually they're dropping with two double hours down, I really need to sort this out now. Like you get so reliant on the fact that that's the that, you know where it's going. Whereas as soon as you start having to fingerprick again, it's like, well, it could actually, I could be like, 3.2 and 5 minutes for all I know where. Whereas when you're on Dexcom you can be like “ohh, great. Yeah. No, I am gonna be that in 5 minutes”. So yeah, I think you definitely get reliant on the fact that. You know what's happening not just at that specific moment in time. And I felt when I used to fingerprick and stuff like I just ended up fingerpricking so much through the day cause you're constantly like “ohh, I should just double check that it's alright”. I just double check that it's alright. I often found like when I was first diagnosed that because you're so used to running high, if I got to like 6, I would feel a bit weird and kind of think I was having a hypo and I'm like constantly having to prick. And like now if that happens like sometimes I'll be like “ohh that feels a bit dry that just cause I've not had a drink in ages or is that cause my bloods are now really high” and I think I'm just very reliant on the fact that you can just know then what it's doing whereas fingerpricking you just can't.

Interviewer

Yeah. Do you think you're always gonna need to carry these spares around or these backup systems or can you imagine a future where that's not needed anymore?

P04

I think I'd like for not to have to carry them around like. Or for them to be so tiny that like it's not a problem to carry them around cause it just ends up. Ohh. I don't really mind carrying a bag everywhere but like I am slightly envious sometimes of people who can let you just go out with a phone and their keys and like that’s it and you don't have to carry around this massive bag full of stuff. Um. I've definitely managed to ween mine down to being a much smaller bag than it used to be, like when you're on pens and things like that. But um yeah, I think it’d be nice if either they get tiny so that you'd kind of don't have to worry about how big they are, or if it's so reliable in the future that you don't have to bother carrying them at all. Or I think the other thing may be is like. So like for the Dexcom, the applicator is huge and actually, if one day it's a really small applicator where you can just carry a spare Dexcom rather than a fingerpricker or because obviously you can't really carry spare Dexcom at the moment because it does just take up most of her bag.

P17

I feel it all the CGM or you like the Libres and stuff, I just got so much like waste with them as well. Like when I went on to the, and even like on the Medtronic when I was changing that pump, the amount of things it comes with. And then on the Omnipod, it’s like flat and like just compact, so I can literally have it in my handbag and it's so neat to put everything together. Whereas like the Libre, the big thing, like the Dexcom, the fact that just goes in the bin. But it's just so everything's so bulky. I think I think things have shrunk a lot since. Like obviously even in the last 16 years of me having it, the things I had, even the glucose readers, that what they're like now. But I think it's still a lot of things you have to carry and even though I've added 16 years, I still walk out the door and forget things. Especially when on the Omnipod now, I've drove like halfway down the road and then realised I have to turn around because I haven't actually got the PDM controller on me. I have to go

back and get it. It's just kind of having that kind of memory to do all these extra things., it's quite annoying.

P04

Yeah.

Interviewer

Yeah. What about people who are being diagnosed or will be diagnosed in next few years? What technology do you think they should be trained with?

P11

I don’t know I kind of feel like it's a bit controversial, but I kind of feel like you want to start out on pens and fingerpricks. Um. Not because I think it's fun. Not because I think it actually works very well. Like it never worked for me, hence why I’m now on a pump. But I think if you don't go through that process. It really worries me if people were to go straight onto a pump, I'm sure it'll be great for them. But, A) there's the physical side, like you were saying, [P04], like when you diagnose your used to being really high and if you go straight onto a system that corrects you back to being in range, you actually feel awful. Like it took me a good few months to get back. I was just like 13 for ages and they deemed that good because it was like not 18. So, I think there's that kind of process. But I also think there's the aspect of kind of learning how serious the condition is, because I think although the technology is making it feel less serious and like yes we can do anything and like doesn't matter if you've got diabetes, you can run a marathon or you know all this kind of stuff. I really like that messaging. But at the same time, I don't want to trivialise it because you know, every year there are people that still die of it because for whatever reason they've either gone too high without access to insulin or too low. And it's like I think you need to learn literally, the fundamentals of how dangerous it can be and how much you have to be able to think about it, so that if the technology doesn't work you have at least have some skills and. But that's a really hard thing to say. Like, I feel horrible saying that because I know how much difference the technology makes to me. But I know that I wouldn't be as grateful as I am for it, had I not had the three years of like horrible in the dealing with the needles and all that kind of stuff.

P17

I think I would have always said go on a pump, but like talking now and like just saying that, you know, when a pump does break and you've gotta go back, even just having a year out of not using injections and fingerpricking. I just think if I'd gone on a pump straight away that, I had like 15 decades or more, I was having like nearly every year growing up. So, I think if I went into one now and I just started on the system, but no experience of that, I think would have been a lot harder because you're coming out of hospital and then you have to go home and get used to kind of being on injections and kind of managing. And like even when like I was on the Medtronic pump and that failed, I ended up in DKA, they were asking me what my like, my daily doses were and I haven't been on injection for two or three years. I didn't have a clue about any of that information. So, I even think

the language that consultants would use in hospital would kind of scare you. And I suppose the people around you as well have kind of understanding and all those things. So, I think definitely injections first. Um. And I never thought about what you said as well about people kind of make, not making diabetes sound quite light hearted condition, but I completely agree with that as well that everyone's kind of making it sound like it's not as bad when we all know it is. It is a horrible condition and you are going to have days where you are booked out and going to all these horrible things. Just try not to like shy away from those completely and make sure that they rather. Don't scare monger people with it, but it is the reality.

P04

Yeah, I agree. I think I was only on injections. I was definitely only fingerpricking for about six months until I got Libre one and definitely less than a year I was already on a pump. But even that small amount of time really made me appreciate the fact that I'm like okay, so I now know how to deal with it if something did go really, really wrong with the pump. And I appreciated the pump a lot more, I think than if you just go straight on it and. Like my I got my pump kind of quite. I was on my pump quite a lot through COVID time and stuff and I kind of ended up just using it as a glorified pen anyway because it wasn't a closed-loop system. And I and then in the moments where it went wrong, kind about having to go back to injecting with a pen and stuff, I think having that knowledge for, if it does go wrong is really really important because technology is not perfect and it probably never will be. So, I think even if it's for a small period, having those skills is really important.

P17

I think what you said about valuing when you pump as well. So, when I was diagnosed, I was always nagging for a pump all the time, because my HbA1c was so high. It was until I'd done DAFNE, that they realised I had the dawn phenomena and they were like the only thing will fix it is a pump. So, I went 12 years and injections of like awful sugars. And then as people were saying earlier because like my sugars were so normal to just reading high on my blood metre all the time, that's why I stopped taking my insulin things. But one of the things they said to me was the pump is a lot more dangerous, so if it does fail, then you know we're gonna have to take you off it because we need to know that you can do it. So, I think if anything messes up on my pump, I go into a lot more panic I think kind of cause I need to get it back on track. So, although it's a bit of a scaremonger to keep your pump and not lose it, it keeps my sugars and range as well. So, I think that works quite a lot.

P04

I think that's the same with like, um, Dexcom stuff. Cause I think originally when we were given Libre on the NHS, it was, if you don't, if it doesn't make your blood better, you will get it taken away because it's not cost effective. And kind of like just that even for me kind of being a teenager was like "ohh okay, I'm gonna actually really try because I don't want this thing taken away from me because even that little step up has really helped me to have a slightly more normal life". Um. So, I think that aspect of it, like however not nice, it is being told like you can have this taken away if you don't use it properly and if it doesn't make a difference, it's kind of like that also means that you do appreciate it much more.

Interviewer

Yeah. What about in terms of, yeah, this kind of like trust side when you do start on new technology, how much would you say you need to understand how the technology works? Or what would you want to know about how it works?

P04

I think. Um. I think like you said, with your Omnipod training, it's like I thought when I had my training for Omnipod, it was a big group of people and you get a big presentation on it, you will do it, try it together. You get a practise one with saline stuff in it. And like I think that's quite a nice way of doing it, cause you kind of can all then help each other. Whereas when I went on my t:slim, I sat in a room with two nurses and they showed me how to use it and then went there were some videos for this as well, but that was also an adult clinic, whereas my omnipotent one was a paediatric clinic. Um. Doesn't necessarily mean I didn't know how to use it, but I was kind of more worried about remembering how to use it because obviously on the paediatric clinic I also had my mum there so she was remember how to do it. Whereas on t:slim kinda thinking to myself, “ohh am I meant to do this first or do I do this first?” Like can't quite remember how to do it and stuff. So, I think definitely making sure that people understand how to use it is really important. Because if you're not using it correctly, then obviously it's not gonna really help. And kind of like when I was on my Omnipod, I kind of didn't get taught how to use the waves and things like that. Like I knew a temp basal, if I reduced it, it will give me less insulin and if I up it, it will give me more. But kind of like the amount of time before you do something you should do that and all of that stuff and the waves and everything, like I didn't get taught how to use. And so actually, then you just end up not using the benefits of the pump. Whereas my t:slim, I've been taught how to use all of that stuff, so I actually try and use it a lot more. Um. But it also might just be where I'm getting older and kind of like caring more about my diabetes, I suppose it's probably the best way you putting it. And I'm actually like, trying to control it. So, if I have pizza I will wave because I know it's gonna go wrong. Whereas if, as when I was a teenager I was like, “ohh, well, my blood's gonna go high. What can I do?” And so I think yeah, being able to teach people properly how to use it will mean that they'll be be able to access more to help them in their care.

P11

Yeah, I think I 100% agree but I think it's a time thing. Like I'm already anxious about it because I've got a two hour appointment. It sounds very similar like a two-hour appointment with one nurse, to move from my Medtronic pump to my t:slim and then I'm supposed to be back at work the next day. And I'm I'm already going nervous about it because I'm like, “this feels like a massive step and it's gonna be completely different”. So, I've already been like watching videos and trying to write down questions because because I want to go into that appointment and make sure I feel like I'm prepared. But then I feel like that's been all on me and I kind of, yeah, I am worried about it because again, they didn't really talk me through my options. So, they just said, well, you can't have the Omnipod 5 cause that's not in your area, so you can have the old Omnipod. And I was like, “well, kind of what's the point”. And then or they would like “or you can have a Tandem, and we think you should have the Tandem”. And I was kind of like “OK”, but then I've had to kind of research it all myself. And it's, it

does feel like again like it's very driven by the funding. And it's kind of like, “OK, so the funding in your area allows you to have this and this”, those are your options, but you're not necessarily choosing the one that's right for you. Or it's only now that I'm preparing to go into that appointment then, like, “ohh I actually feel really underprepared”. And it's not that I regret my decision. But there is also the fear of like, “OK, well, if it doesn't work, if they really don't get on with it. Like what happens? Can I go back to my old pump?”. Well no, so I'm stuck with this one for four years. Like it's a real, I don't know. It's it's a real worry. But I'm sure it would be fine, I’ve heard really good things. But it's just, there's not a lot of support and not not a lot of, like emotional support either. I feel like it's all very practical, like, this is how it works, this is how you set it up, this is how you fill it with insulin and go. But actually, that whole anxiety thing that we were just talking about with new technology and dealing with all the alarms and the noise and like the the sense of actually being responsible for your own care suddenly, in a whole different way. No one really talks about that side of it with you. So. Um, yeah, I don't know. I miss, when I was at university, we had a group of diabetics at university and that was the best thing because it was like we just had a group chat and ever messaged each other being like “I'm having an issue with this. Or does anyone have a spare this?” And I think that's what people really need in a way, especially with all the new technology is just a kind of group form sometimes just to be like, I'm struggling with this. Because actually also the other thing is the nurses don't always know. And my nurses are lovely, but none of them are diabetic. So, they know all the theory, but they've never actually had to implement it. So, actually there there's a limit to how they can help sometimes I feel. And whereas sometimes I just wanna ask someone else who's doing it everyday.

P17

I think, going back to you said about the two hour training, I think when they put on there, that it was gonna be that long. I knew there's gonna be other people there just because I had spoken to them on the phone a couple of days ago. But when I got up there, it was a bit of a weird one for my. I’m in adult clinic and I have a much better experience of adult clinic than I did paediatrics, which I seemed to find any kind of engaged in any type of conversation and the only person with that experience. But, just looking at when I went into that appointment I was one of the first ones in the hospital, there was four of us. We were the first forgot Omnipod 5. So, their initial training session we were doing was the first time that the nurses were doing, so they were asking a lot of questions um so they could learn. And at the beginning of this is really reassuring, they're learning, but then the back of my head was like, well, if they don't know how to do anything, how am I supposed to ask them. Because they were taking notes because they were gonna deliver the session themselves going forward. But even when they ring me weekly, they we have like a group conversation, but they are really good, they'll be on the phone for half hour with me going through my data and then like pinpointing things. Um. So, I've got a really good relationship with my team but I think like you said with the forum and having like some sort of conversation with different people, I just turned to Instagram and ask everything on there. Like the diabetic community on Instagram is like my go to, I'll just put it on my story because I've got an account for like my diabetes and like, I'll get a response instantly, like, “phone this person, I've seen this post”. Um, and I kind of go through that and I think when I first kind of made that account, I was coming out of like my kind of anti-diabetes and I didn't want to get to know anything when I realised there was other people who experience the same things. And I think if I had that kind of support during the appointment and different things. Like using that kind of psychological side and support, it would have been a lot easier going on to different things.

Interviewer

Yeah, yeah, I guess the other side of this is like technology is getting increasingly more complex as well. I guess there's still in the realm of like the closed-loops kind of makes sense that if you're blood glucose are going up, it will increase the insulin. If they're going down, it will decrease it. But we were starting to talk about in the other interviews, the increased use of artificial intelligence and more complex algorithms within it to the point where it might be difficult to really understand why it's making decisions it is. How do you feel about that as a piece of technology and what would kind of make you trust it more or help you get used to using it?

P17

I think the algorithms is something I'm quite fixated on with the Omnipod. It's just kind of watching it. And I think now I've been on it while I kind of understand what it's doing, I can see, I can think back in my day. So, what I've been doing on the Glooko app is a really good like feature and you can add notes into different things you've done for the day. So, if I know my sugars have gone up, I'll put a note on there and be like, "ohh, I went for a walk or I give my insulin a little bit later after my lunch", just so I'm making mental logs. Just so, if I'm on the phone to the team then and they're asking, they will ask you, "what were you doing Tuesday at 6:00?" And we have no idea, I can't remember it in this morning, so I can go on there and look through it. But, I think the algorithm is really interesting that how it kind of registers everything. I don't think, I've put a lot of trust in it, but I don't think I'm 100% got my trust in it. I just think if something goes wrong or like even the battery life on my PDM for my Omnipod worries me. Because a couple of times, like this was the last five days I have to charge it every two days. And I think even though was like a USB C charger which is quite universal, it won't work on any other charger. So, if I haven't got that charger with me, I panic and I'm like "right, so I'm going to have to make sure that's in my kit bag now". And now that everything else is condensed I have to take that charger. So, I think what I haven't got the algorithm there, I kind of got into a bit of a panic kind of come back to let that watch thing as well.

P04

Yeah. I think going from the old Omnipod PDM that had batteries in it, so I could just can't carry a couple of batteries with me and it would alarm when it needed changing and stuff, to having one that I basically is attached to me and then I have to attach it to a plug and charge it. It's kind of quite. It kind of takes time out of your day. So, then I'm like, "okay well, I won't do it everyday", and then suddenly I'm at 45% and then I'm like "oh my God, it could stop at any moment", even though obviously it's not going to. But like I I get quite stressed about like being like "OK, it's slightly low on the charge now I'm gonna have to sit for ages that are plug" and then you can't really do anything in that time because you're attached to that. It's not like I can take it off and walk around or anything. So, I think that aspect of it is slightly annoying, so it would be good if there was a way of like not having to be plugged up to a wall er for half an hour. But, that's obviously something, but we'll just come with time.

P17

And I think the algorithm thing, like the last couple of days, I've had two or three days now when my sugars have been like 98 and 99% in target, which blows my mind. But because I'm seeing that data on there, then I'm like, what have I done in that day. And then cause I know I've got like a busier day, so it would been days where I've just been working at home sat the laptop, take the dog for walk, go to the gym. So, it's kind of just normal things I'm doing. But when I know I'm in work then with that young people and can be running around doing stuff, I know that algorithm is gonna change. So, in my head then like I'm kind of fixating on like the data rather than actually. It doesn't matter if they're line has gone out that this still in range. So, I kind of get fixated on the data and which I think sometimes like I said, everything I said in my interview is I look at it too much, but actually just for getting that I need to live around my diabetes data as well.

P11

Yeah. I think the thing that worries me is always that kind of like you're saying is like everyday is different and there are a million and one different variables. And so like the type of exercise I'm doing or if I'm on my period or what I've eaten or how well I've slept. And like, I feel like for an algorithm to be able to cope with all that, that would be amazing, but I somehow don't trust like it will. And I still feel like, you know, we still have to put in our boluses um, so there's always error, there's human error. Because like, well, I look at plate of food and if I cooked it myself, I'd normally weigh my pasta but like, sometimes they don't um or if I've made a sauce. Like tonight I'm supposed to have fish pie whenever, I'll probably weigh the potatoes. Will I weigh the amount of flour I actually put the sauce? Probably not. So there's, like, this kind of there is a human error aspect and I don't think you'll ever get rid of that. And the amount of variables that happened in the day is a lot. And I think also the other thing that's worrying me about changing the system is like learning how different systems work. I think you do get a bit of like mental overload that. I kind of learn how the Libre worked fine, then moved on to Dexcom, moved on to the Medtronic. And I've got my ways of dealing with Medtronic and it's it's imperfect, but it works me. Like, I've got temporary basal, but I don't think you can do that on the Tandem with a control, you have to put on like an activity. So, it's like it's already overwhelming me and then again, having to go back to work the next day and you're like, I'm really worried about "OK, well, then what happens?" And if I'm trying to teach a class and then suddenly my pumps done something that I've not told it to do. I'm used to having a pump that I tell it to do everything and I've worked out the way that it does it. So it's just. Yeah. Again, it's an anxiety thing. I think once you're in it for a while, but it's the first month or two that you're getting used to it and that's. It's really scary, especially when you've got like responsibilities. I kind of wish I'd been able to do the beginning with the summer holidays but like you don't get that autonomy of choice and when you move on to stuff. And it's hard to reconcile like your life with all the changes.

P04

Yeah, I agree. When I went on to my t:slim, they wanted to do it when I was on a placement on studying physio and so it's quite a, it's quite busy and I'm used to my Omnipod, just like doing a temp basal for the whole day cause I know I'm gonna be walking around all day. And kind of like they were like "ohh we can get it done for you this day" and I'm like "cool that's great but actually I have no idea how to use it and I'm gonna have to, I'm pretending to be a physio and I don't want it to affect what I'm doing". And so I put it off until I wasn't doing anything but then I was still at uni doing lectures and stuff and I study in [UK city] and it's just hills. And it's kind of like, "ohh now I don't have

a temp basal setting, I've just got an activity setting. I don't know how long I need to put that on before I start walking" and things like that, so it's definitely. But when I did my Omnipod thing, cause it was going from pens to pump, you have to get up in the night to test your blood sugars because I didn't have a Libre at that point. And so, I was allowed time off school, I was like that's perfect. But I can't just get time off anymore because I'm not 16 years old. So, it's kind of like, but it's a very big adjustment and they kind of don't explain to you how it's gonna change. But then now I've got used to it, I can't ever imagine going back to not having a closed-loop. And when my closed-loop, because sometimes my t:slim will for some reason just not be connected to my Dexcom and it will just go back to its normal basal, I'm I find it really stressful cause I'm like I'm so used to using that closed-loop system now that now I'm back on a normal basis. I'm like "ohh is it gonna go really wrong in a minute?" And yeah, but it's it's definitely an adjustment but it's definitely worth it.

P17

I was gonna say it's a big mental shift. Like you said, you're so routine to doing things. And like you said with the Medtronic, I was really bad for like rage bolusing all the time. Just being like, yeah, rather than running correction dose in, I knew how much insulin I'd need. So, I'd be like "ohh, that's 20 carbs", because I knew 20 carbs was a certain amount of insulin and like my team had never picked up on that. But just as I spoke about that like on Instagram knew other people were doing it. Obviously, we go bad habit and there's as soon as they went Omnipod, I had to completely iron out all these bad habits because it wasn't the way with like you said with the activity mode. It's like I was going to the gym, they were like "you need to, if you go for an hour, you need to put the activity mode on an hour before you go and leave on for an hour afterwards". And I was like, I'm not that organised to know that I'm gonna be going to the gym in an hours time. It's kind of like whatever time I finish work or if I'm gonna be in traffic or something. Um. I think that was quite difficult adjusting and realising that I kind of need a routine. But one of the good things I think on the Omnipod that with the routine is like you change your pump at a certain time. Whereas, with that Medtronic it was whenever it ran out, with Omnipod it was every two days. Um. But one of the other things I found was like the algorithm is it kind of would predict different things. You could almost see your trends of where it was going seems gonna read. So, I was kind of looking at that and seeing where I would be with things. So, yeah I think there's a lot of pros and cons to it but I think you get used to it and once it works it feels like you can't live without it. So, like my sugars went to 13 the day before yesterday and I felt awful, and normally I could 13 would be pretty good for me. But because I'm I sit at like 7 like flat lines everyday this week it's it's mad. But then the thing, the other point you said as well about like the appointments when you go for that two hours and you're left. I took annual leave for that day just so I could like get used to doing it, but they were like kind of stay getting used to your daily routine and then I was thinking this is the busiest month me I've taken like my youth work group away for a week to do outdoor pursuits, so I'm going to have to wear the Omnipod there, getting used to the Omnipod falling off all the time. And it was kind of like managing 10 young people, who've got learning difficulties as well and kind of managing everything around me. And it was and then they were like seeing how it was with alcohol like any kind of lifestyle change I've experienced over the last month. So, I feel like I've been, it's been really overwhelming, but I know it's gonna be good so.

Interviewer

Yeah. I mean, it seems you're already highlighting the importance of this kind of onboarding process and what needs to be there, particularly with new technology. I guess maybe moving on a little bit, I want to kind of think, yeah, more into the future and kind of how you think that technology is going to shift or where you can see changes coming in future.

P17

I just hope is more accessible. Um, I think we like the postcode lottery thing um I find that it's quite difficult. So, I was quite lucky that my pump broke and when I rang that the warranty run out. No one understands why but there was like 7-8 months earlier and I've just fell in line with going on Omnipod 5 where it would have been waiting and they were gonna ask me to go back on injections until next February when they should have been having a pump. Um. So, I think funding and different things. And just think back to you know the I know like the cost of DKA in hospital is a couple of grand. But, if you put me on a pump that's controlled, all my sugar, is it, you know, it would, it's probably like only like 10% of the amount of DKAs I've have had. So, it's kind of that cost effect and per and trust in us to that we manage our condition. But if we have a lot more support maybe the psychological side there's more investment in that side then it will reduce things and technology might become more available.

P11

Yeah, I think maybe just a bit more regular support or. I kind of feel like I'm satisfied with where the technology, sounds bad, but kind of where we're at all like. Especially with the if we can work out ways to integrate this data that you're looking at in terms of smartwatch stuff and like. I feel like we kind of that data peak, I'm not sure I want more data about me, like more things monitoring me. Or like you know I think we probably could go to level where there something inside of us monitoring, and I kind of don't want that. I'm kind of happy with what I've got cause I feel like it, we're at a point where there's like the balance between the AI kind of control and the balance between the human input. But I think what's not there is the the human support to deal with all that technology. Like you think about the last five years that I've been diagnosed, started on pen, Libre 1, then Libre 2, then Dexcom, pump, pump 2 and now pump 3 in five years. It's like that's a lot. And I I kind of. You kind of want more regular support check in. And they they made me do DAFNE for their first pump I went on and that was like four weeks of mega intense information. And I learned it all for that pump, and now I probably need to go back and do it all for the for the new pump and like. But it then all falls on you and it's just there's just so much that you can think about and I think probably my diabetes care has slipped over the last year because I'm almost in like overload of information and there's so many things I could change. Like based on what they've told me and based on the Dexcom data and based on what my pump can do. I don't know where to start and I I don't know what's most important and then I slip into the bad habits of like rage bolusing or what I was doing on holiday was just putting a temporary basal on for the whole day. And then I was going up to 13 and I was like, well, this is fine because it's a lot 3. But actually, I probably didn't need it on that whole day, but you know it's just like there's not the regular kind of support. I almost want, sounds really bad, but I almost want like a weekly half hour, where we focus on something for a week, just something tiny. But instead, I get one one hour appointment once a year and then a two hour appointment when I change pump. And the rest of the time it's like on me and that I don't know. So, I think, yeah, that's really where the investment needs to be. Or in trying to connect other diabetics and realising that actually there's a

lot of group knowledge between u. And especially amongst young people I found that I have spoken to you like who are trying different apps and trying different technologies and like. We're the ones who are living with their day-to-day so yeah, I think help us, but also connect us and maybe group learn as much as someone without diabetes, whose 51 telling me what happens when I drink and they have no idea what I drink and when. You know, all that kind of stuff, so it's just yes, the human side I think they're really needs developing at the same pace as the technological side.

P04

Yeah, I agree. I think I had when I was first diagnosed. I think it few months after I had one like psychological appointment with a therapist. But it's kind of at that point I was so like not understanding what had actually happened, I was kind of like “yeah, I'm alright” and they were like “cool” and then like that was it. But I think we are quite lucky cause er I didn't get much support through COVID cause I was meant to be transitioning to an adult clinic. Um. I had one appointment with an adult consultant and I hated her. Because she she just liked, didn't seem like she cared, she told me she, like, looked at me and went “ohh we're gonna get your thyroid checked because it looks a bit prominent”. I was like, great, wonderful, cause I'm celiac as well, I was like, “great, all three. I'm doing really well”. And and it wasn't, it was just cause I've got a skinny neck. But she likes the way she said it terrified me and I was like, “right now, I've got three conditions to deal with”. And so when I moved to uni, I decided to move my care down to [UK city], cause obviously that's one of the leading places in the country and [details on employee of the hospital] and they've got a young adult clinic. And so, I get, I still get three, I get appointments every three months still. I obviously have a gastro one once a year. But I understand your pain cause having a gastro one once a year is annoying. So, I can imagine having a diabetes one once a year is a complete pain. But yeah, unfortunately I'm quite lucky I get I still get quite a lot of support. But I think like that therapy wise, it might be quite good for us to be able to quickly access it if it was needed, cause it has been times before that I'm being really sick of it and I actually it'll be really nice to be able to talk to someone and go “I hate this at the moment”. Because I don't really know anyone who is diabetic, like there's there's no one at uni I know that's diabetic, no one at school was. So, kind of didn't actually get to speak to anyone about how I'm feeling. Um. So, I think that aspect of it be quite nice. And then I think technology doesn't necessarily need to get better, I think it probably just needs to get a bit more streamlined. So, kind of like being able to use the wearables and so with machine learning kind of like it learns your pattern rather than this set one that everyone ends up using. Like the activity mode is like it's just activity mode, you can't change it. You have to remember to put it on stuff, but actually if having a wearable means that it would be able to work out how that works for you and how it works best for you. And actually, ohh your activity mode currently is connected, meaning your corrected to 7, but actually yours needs to be 8 because your bloods are a bit more reactive and it can change that for you, whereas you can't do that now. And then I think silly things like, cause I'm on a t:slim, it's obviously attached, it's got a clip. And so, if I clip it under a dress, I then can't get to it easily without having to hoick up a dress in public. And I think in America, they've got an app for t:slim, so you can do it from your phone, which obviously would be nice. Like that sort of thing, kind of like making it so the technology we've got is just more accessible to use because otherwise, like I'm having to hoick up my dress in public or going to the toilets to do my insulin, which is something that I've never wanted to do and never have done because it kind of makes it more taboo than it needs to be. But obviously I don't want to like have to bring up a dress in public just to get to my pump. So, I think kind of making it more accessible is kind of where it needs to go rather than necessarily making it so it fits inside your body. However, that might be cool. I don't really know.

P17

I think the accessible thing you said is quite thing, because I think for me one of the best things about gonna going onto an Omnipod was that it's not, even though it’s attached to me all the time, it’s out of my way. So, I can just pick up my controller, it looks like a phone, so it kind of removes that taboo with it. But again, it's kind of remembering to take that. But with the Medtronic pump that clip used to break probably so many times I'd have to order a new because I would break so many times that anytime I ordered a new my three month supply would come through I's been asking them to put two in there and they'd be like “we can't give you two”, but I'm like, “they’re breaking” and then you speak to like in your team and they be like “ohh, they shouldn't be breaking that easy”. But you don't wear ones you don't know. Like I'm only clipping it onto my jeans, but trying to get it and put it in certain places. But I think the only thing with the Omnipod now is it's gotta be in line. So, whatever your Dexcom is, it has to be in line. So, I find it difficult that my Dexcon will last 10 days, but then change my Omnipod every two days. So, it needs to be in line of sight of where it is. So, they were saying like, you can't have it. So I used to wear my Dexcom on the back of my arm and I was like, brilliant I can wear them both. But they were like because of the Bluetooth, you can't have something in between, so I can't win them both. But I think even like how devices look as well, I think things are better and there's companies that do kind of like jazzy stickers and like adhesive kind of patches to keep them on. I think things like that should be available, whether they're available on prescription, I don't know. Or even promoted in clinics, if you don't know about these things. It was only through social media, I found out about them. And I mean some days I don't really want my tech on show and I'm kind of quite happy with it looking plain. But other days, I'm like ohh no okay I quite like that you know. Especially as I work with young people, that have tech and things themselves, there's a diabetic. I quite like having it that it matches something. So yeah, I definitely agree with the accessibility.

Interviewer

Yeah, I guess there were kind of a couple of mentions like partly to do with this study around yeah, having the smartwatch and feeding and data from that. Um. There is a lot of information that comes from the smartwatch are there particular times you can see that being more helpful than others?

P11

I definitely feel like the activity side cause mine seems to be super sensitive, like even if I don't tell it that I'm going for a run or something, it's picking up, which is amazing. Um. But, I'm still not sure. I don't feel like I'm using the Fitbit to the best of its ability because I'm like, getting all the data from it, getting all the data from Dexcom but I don't know how to put them together yet. So I think, yeah, definitely people like go in between so you can suggest stuff like and you can pick up trends because. For me, I'm not sure the Fitbit yet, maybe I haven't turned the setting, but it's it it's not telling me trends. It's not telling me like you often go for a run at this time and it's not doing that thing where it can say like when you're around your blood sugars like. So, it's trying to integrate the two cause I feel like now I've got a lot of information like could do stuff with. I think especially for me things like menstrual cycles and things like that, I found it really useful being able to put that information into the app, the Fitbit app and like have it all streamlined. But again, I don't really know what to do with

it yet, so it's. I can see it being really helpful in those kind of ways the physical things that we doing with our bodies that are gonna affect your blood sugars, but I think there does need to be some guidance or support in dealing with that data and making suggestions. Because the amount of information overload that we're already dealing with, to then get more data and then make a informed decision about what to do with that. Especially if your pump can't change its activity mode or whatever it is like. Someone to help with that would be really would be really good for me, I think.

P17

I think the data on the watch, like obviously like even like the sleep patterns and different things. I think if you can consultant team can see that it almost paints more of a picture of you as a person. So, when you've got there. And I think my team are quite good they will ask like what type of exercise I do. So, I know when my HbA1c came down through lockdown when I started my pump they were like “is there anything else you’re doing?” And I was doing a lot of HIIT and home workouts and they were like “keep doing that because that's working”. And then as we came out and went back to the gym was doing strength based and they were like “right, we can see the difference”. So it was having those types of conversations. And I think I think of a lot of other people could have with conversation even like seeing the sleep patterns. So, like you were saying, you were waking up at certain time of night. If they've got all their data and they can see right okay that certain times you'll waken up, they could analyse that and kind of take that thinking away from us. Um. I think that would be really interesting to see how they kind of take it and a make suggestions, would be really good. Just to paint a picture of you as a person, rather than just data from you.

P11

I think what you said, the big thing, like is taking the thought process away from us. I think that's the major thing for me. Like, I really like having all the data, but selfishly, I don't wanna have to be the one that sifts through it all in. So that's the thing for me like, I think it could be really helpful if they're the ones to then act on that.

P04

Yeah, I agree. I I mean, there's times where I've been suggested things by like, the diabetes nurse that I've been like “no, I don't agree with”. As like when I first started my t:slim I kept going low a lot. Um. And it was because um, my thing was set so that I had slightly too much, and then they were like “ohh, maybe change your basal” and I'm thinking “ohh no, but it's after I eat so surely that's my bolus”. And when I used to be on my Omnipod, when I first set it up, I did it so that it would, when I did boluses and stuff, it would correct and correctly to 5.5, which I quite liked, cause I run quite high all the time. And then when I'd go low often they're kind of like, “ohh well maybe up it to 10” and I'm thinking “ohh no, but ten that's quite high, I don't want it to correct me to 10”. So, like I'll try 7 and then. So, like I kind of like thinking having that extra help is good cause then I can try and make my own decisions on it for when I probably won't sometimes have as much access to the healthcare professionals as I do now. But I think that's probably just a personal thing because I do get quite a lot of input as it is now, so I need to learn how to do it myself for those times where I'm too old and I'm not allowed that input anymore. So.

Interviewer

Yeah, it seems to be we're talking about this kind of advice coming from healthcare professionals. How how do you feel about it coming more from, I mean even algorithms, if there are some kind of trends, it's noticing having access to it, would you still feel able to use that or would you kind of trust it less or yeah, how you would feel generally?

P17

I think just going back to what I said like with the health trends, I said at the beginning being like “ohh your sugars have been this doing it”. I think that is quite useful seeing it and be like as a positive thing “ohh your sugars have been range”. I don't know how I'd react to the trend if my sugar weren't as good as they are at the moment. So, if they would come back being high like even just on my general fitness things and if it says that I've been like even like the circles might watch if my don't hit my stand goal and I can see that coming down I get annoyed because I just saw competitive with myself. And there was a really good app um that someone on Instagram had messaged me about and it was to do with algorithms with the Dexcom. So, they created an app that linked to Dexcom for young people that would read your Dexcom data and then it would literally, it was called Nudge. It would literally send you a nudge on a video or a post relating to the things you were doing. So, like if you put on there, I don't know, you were gonna drinking or, it would send you a video with that and it would send like helpful information. So, maybe something. I know they were trying to buy into like the NHS, but something like that. But again, everything's always targeted at children and young people, and I think sometimes we forget that adult might need it more. Because you get so much support, I think, as young people and and as children. But as you go into like adult, I've just, well, I'm 26 now and I think all these changes in technologies are still really overwhelming and all this data coming out. So maybe it. I think they could be really helpful if they invest a lot of time in putting the educational side and the emotional side alongside it.

P11

I think that's true actually, like if an algorithm was able to say, “OK, we've analysed the data and we recommend that I don't you put your activity mode on half an hour before you're currently doing”. But then I think you would need to explain it but like “because we think this will make an impact”. And like I think you need to be able to understand the impact that the recommendation is having. Like I think if someone, even if my nurse like it's the same as if she says “just change this”, I'm like “well why, what's that gonna”. And you know, so I think it's their understanding, I, still it comes back to the anxiety of if something breaks because I still want to know what what is going on, so that guess if if something was not to happen, I was, I would be able to replicate it as best I was able to. I think I still want some level of control or knowledge input in there, if I can.

P04

Yeah, I think I agree with that. I think I would want it more so that you can make an informed decision based on what it's telling you rather than it automatically telling changing it itself, because obviously that then is quite stressful. You're like “Oh my God, why is it changed my carb ratio?

Because I've been on that one for three years". I think it's much more of a like a using it for informed decisions rather than it automatically doing it for you. Because even though that's easy, that's probably quite stressful for a lot of people, so.

P17

I think that's a really good thing I said about the informing what you've done it because I've noticed on like the Omnipot when I've looked like how much insulin I've used, I think my total daily dose used to be like 150 and since I've been on the pod it's been on like 49 a day. But I don't understand why those things are coming down. Is it because I am being more routined and I've kind of gone back to being way my food and really checking the things overly? But I know realistically in a couple of weeks time I'm not gonna be that consistent because I know I'm gonna start slipping back into kind of being relaxed with things. But if you could, like, notify you be like, "this change was made". And even though, like the Dexcom Clarity app where if you look at the trends, you'll see all these patterns, maybe look at your bolus or something like that. It's kind of talking points that with your team. So, I know I can contact my team quite easily on the phone, but I know other people can't do that. So, I think there's a lot of changes that can be made within technology, but it needs to line up with everything else. Because, I think if all these kind of like you know like Dexcom, Medtronic and start making these changes with that algorithms things, which are all amazing, but then the them got the staff and the knowledge on the ground to kind of support it as well in the teams. Um. I don't think it'll be as beneficial to as many people as they think.

Interviewer

Yeah, it's interesting you saying that you kind of want you be kind of receptive to it as an input, but you'd still wants to be making the final decision yourself. Um, do you think there could ever be an algorithm that would be? Yeah, the better kind of making that decisions or like suggestions than you would be?

P11

Maybe. Like maybe. I think it's hard when it doesn't exist and I think it for me it comes back to like what we are saying about that it is a really challenging condition. And I think so much is going into it to make it feel less of a burden and like it's really good. And if something could do that and yeah, it's amazing. But I just don't feel like at the end of the day, it's my body and I know that I could die from this and like, that sounds really awful, but. So, therefore it's a massive decision to to not have any input into the decisions it's making it. It isn't just a kind of "okay, what colour is it gonna dye my hair today?" It's like I could I can die within a day probably if something goes wrong or in a few hours even. So, if I'm like on the top of the mountain and it decides to do something like and I don't have the resources or the people around to help. Like I don't know it's, it's not just a yeah, an AI that can choose my outfit or something like it's a life and death thing. And I think there's nothing that exists at the moment that is on that serious level and therefore I don't trust at the moment because it's it's such a big thing. It's not like a feel like a lot of AI out there at the moment is quite, well that's publicly accessible, it's quite fun, it's quite like helps you with your lifestyle or like helps your work and like, but nothing that actually impacts your life and death. And like that's the shift that feels quite big and just too much for me to think about the moment. Like, I don't trust. But maybe if I could see I I

wouldn't volunteer myself to be the first person. I'll just put it that out there. But like, if I could see it was working, then maybe.

P17

I think one thing I've just thought about, as you said that then is with the AI think especially with like I know when I was added on to like Snapchat and everyone was using it and it was kind of like this whole thing that people were using it. And like ChatGPT, everyone's quite anxious about what it can do and things, but also how helpful it is and that it can do a lot of things. And I think if people think that if technology becomes so good that and everyone kind of talks about “oh AI is kind of in control of my diabetes”, I don't want people thinking “OK, diabetes is a walk in the park because it is doing everything for me”, when actually there's still a lot of things that you need to be doing yourself. Like even when they're my team were trying to like, sell me the idea of Omnipod, even though I was really keen and I really wanted to go on it, they were selling it to me as “well, you won't have to do this. You won't have to do that.” And even though all these bad habits and all these things I've learned to trust it with, there's a whole new system and new things I now need to learn and do, so they're just replacing other things as well. Um, so I just wouldn't want people to think “ohh yeah, diabetes is”. Ohh like people say “ohh you have an injection and you're fine”, when we know that's not the reality.

P04

Yeah, I think I’d be much more likely to trust it if it explains why it's making changes to things. Or like I think also like having an option to override what it's done in case it was to go wrong. It's kind of like even like with carb counting, it's. Like sometimes I'll my bloods will be slightly like they'll be like 5.4 and then I wanna eat chocolate bar and I put in the carbs and it says that I need no insulin for it cause I've got too much on board and I'm like, “well, I know that that's gonna make it go high”. Um. So, I can override that, but I I think if you have no option to override it, then that's when it could potentially become dangerous. And I don't know how it would work, but like it is technology people can hack into technology and things and so like making sure that it's secure and safe enough that those sorts of things couldn't happen because obviously it is kind of a life or death device um and it could end up being used in a malicious way if it's not, got those security measures in place. So, I think things like that are probably be really important, being able to override things it does if you don't agree with it. And provided, I think lots of people probably be fine using it, provided it explains why it's doing things because like if it's just like “ohh I've changed this” and you don't know why but then that doesn't feel like you can trust it in the same way as when you're working those things out yourself. Um. And I think things like, it's really hard because you kind of just get a set amount of insulin for the amount of carbs you're having, but actually all carbs are different and all carbs affect blood sugar is in a different way and so it probably would be good for it to be able to learn how different carbs affect your body, and so if you can put in that you're eating pizza and it will know that you'll go low and then high afterwards because of the fat. And it can help do that for you rather than you having to do this whole trial and error thing of working out how to put in your insulin for meals like that. I think they probably be useful to make life easier, but only if it's used in a safe way.

P11

I think as well, like it's all about the routine. I think if it's if something's done really repetitively, it'll it'll be able to pick up on that. But I know my days like I can know what my timetable is gonna be like and then I have a kid walking and cry on me and suddenly the whole that you need, like suddenly I've gotta take one kid to the the health centre and like another kid's going home and you're like. I end up running and how could a piece of technology I don't understand at the moment how that could ever respond to those quick-fire changes that I'm dealing with. And like one day I might make a pizza with loads of cheese and then the next day I'm like ohh add some ham and then that's more protein. Like I don't think even even the pizza can be different on it different day and that you can add more fat or less fat and like if you go out to eat you don't know what's in something. And like there's just so many variables I think it's just as a lot and I kind of hope one day there might be something. Like maybe you could scan a picture of your plate or something and it would be able to analyse it. But until something is fully able to cope with all the variables it doesn't feel, worth it is the wrong word, but like. I don't know. If it can only cope with certain number of variables then it's always imperfect and so there's always going to be more that you have to input into it. I don't know., I think it's baby steps, maybe. And I think for me something that's able to deal with like activity is a massive thing. So, like if I go for a run or if just now I log off this call up what 4:00 and I'm like actually I fancy going for a swim, I can't do that at the moment because I haven't put a temporary things on. But like if something could respond to that in like helpful lifestyle stuff, then I'd be, I think that's a big thing for me. But I don't think it could cope with all the food and like. I don't know the actual amount that we have to deal with I think we don't realise how much we think about on a day-to-day basis that this would also have to think about.

P04

Yeah, I think unfortunately. Technology is never going to be able to mimic what's happening in a normal pancreas, and I think that's probably why they are testing things like genetically engineered beta cells and things like that, that the immune system won't attack. Because actually, like you said, the technology will never be able to deal with a crazy lifestyle. And but, yeah, if you're sitting down every single day at a desk and eating the exact same meal every single day and stuff, yeah, it might be able to cope with that, but it won't be able to cope with a normal life at all like a pancreas would.

P17

I think, like you said algorithms are not built for if you're spontaneous or an organised like me. Like and my day is never the same. I can try and plan and be as routine and organised as I can. And like I mean everything's written down or it doesn't happen and still things change and what you say, if you're on your period, your diet, it could be absolutely anything, even sickness. Like that's one of them really anxious about is when I have to apply sick day rules on this pump because I have, even though the regular just following the same way. Well, actually I don't have temp basals anymore, so do I put activity mode on for that, things like that. And I think it's kind of that I like having control of what I'm doing, but I think yeah, it can kind of. Yeah, I think is really good at the things that are happening, but there's still a lot of anxiety around what's going on. But I think it's really good that, you know, there is small changes. Even like things bringing in the smartwatches and kind of if they were introduced like when you have a pump would be like recommended or even. I know they wouldn't fund a watch on the NHS probably. Or even saying like they had like a deal with Fitbit or something and being like, so I know Diabetes UK have got one for like the walking challenge being

like, right we we recommend you get this because it will help you and understanding why would be really helpful. So, they could give a discount code out or something. But it's just something just kind of feel a bit better about yourself when you're diagnosed. Cause I think you're given all this kind of you know you've got this massive burden to carry for the rest of your life and I think. I don't know how it's different to when I was diagnosed like I had a fit and I woke up and they told me that like, I can't go in the army, I can't do this, and you're you're gonna you're stuck with it for the rest of your life, your thyroid doesn't work either and it's kind of like a lot of information just digesting about the condition. So, if they can kind of help with giving almost perks to the condition as well and saying these are good things, because I think as much as I complain about the algorithm and different things, it's definitely help me in, I know if I'm going up for food or something with my friends, I can plan my time around it. And I can't think of anything off the top of my head, but I know there's been perks where even like get into festivals and different things you obviously get tickets cheaper and things. But if I'm planning my days around things like with my friends or I'm going to the gym they where I know I'm gonna get tired, like look at my watch and go right well I know I need to eat now because my sugars are dropping. So, I think there's definitely like perks. And I think another thing is quite interesting is seeing the fitness industry allowing people to just buy like Libres like call different things. And my Facebook is full of like adverts for them and my friends just tag me in them being like "oh, they shouldn't be wearing a Libre", and I'm like but it's just reading data in a different way of. But obviously CGM is just obviously branded for diabetics, so it would be interesting to see as diabetic technology advances with like the fitness industry and the food industry following the same way cause it's almost like they're step behind what we're doing.

Interviewer

Yeah. How do people feel about the sensors now being marketed more to wider audiences?

P04

I think it actually probably helps people to realise what diabetics do on a day-to-day basis. Because actually like, kind of, when I was diagnosed like like no one had a Libre, no one had a Dexcom and stuff, everyone was finger pricking and you kind of like, think that you're the only one that exists. And then now you can go and walk down the street. You're like, "ohh, they're wearing a Libre, ohh, they're wearing a Dexcom, ohh, it's not just me". And kind of like other people now being like "ohh, they've got something on their arm" like I've had like so many people I know asking me what does what and "ohh what So what does that one do that's on your arm", "what does what's that that you've just got out" and things like that. And then they'll see other people wearing them and kind of know what that means and what they have to deal with on a day-to-day. So, I think it's probably making it more of a high profile thing, especially we're kind of there seems to be a lot more celebrities and stuff now who kind of wear CGMs openly and kind of market that they're diabetic and that's something they have to deal with and kind of work with the charities and stuff. So, I think like it, provided its not kind of made as a "ohh this is a really this is a device that makes their life really, really simple and easy and there's no now no worries to their day to day", I think it's fine because actually it probably does raise that bit more of awareness cause people be like "ohh isn't that for diabetics" and kind of that sort of thing. So, I think it's probably something that's kind of a little bit of a tightrope thing. But I don't think there's anything wrong with people using it. Because like normal person, normal, normal people, people with working pancreases, their blood sugars fluctuate

anyway. So, if they want to be able to manage that and the kind of look at that, then I don't think that's necessarily a problem with that, provided it doesn't, then make diabetes look easy.

P17

I think it's interesting. I always called like the common section when I see these posts to see, like, the debates come in the comments. But there was there was one the other day that someone had commented and there was like everyone kind of like reacted to exactly what you just said. But then someone said if you had an accident and you were found unconscious some way, they people would know, diabetics of where Libre is or some sort of device you'd be looking for it. So, if they become mainstream, would people not think about that? And I never really thought about that before. And then another one I seen there was an article where rugby player was wearing a Libre and basically doing exactly what we're doing, analysing how exercise affects, you know, while wearing it in different things. And then people were commenting that I think at the time there was a shortage on Libres and and different coming from like I think through Brexit and different things. And they were saying if they are getting mainstreamed, does it mean that there's gonna be harder for those who actually need it for a medical purpose to to get hold of those devices. So, I think again, there's a lot of pros and cons. It's interesting seeing more people, I think talking about diabetes technology without knowing his diabetic technology though.

P11

Yeah, I was going to say the thing he said about marketing cause there's some I see. I think it's called like NOMO or it begins with an N or something that's popped up on my Facebook. And the advert for it because I was interested because I was like, “that’s a Libre”. And the advert for it was like “I've noticed that when I eat bread, my blood sugar spikes, so now I've cut bread out of my diet” and I was like, that's that wasn't helpful for me because they were using it as a way to like target foods and weight loss and like “I've been able to lose later weight by cutting out these foods from my diet because I know that they spike my blood sugar”. And I'm like, well, that's. I don't want people to then turn around and go “ohh OK, so you had a pizza, that's so that spiked your blood sugar, so you shouldn't be eating pizza”. Like I don't want it to then become another way for people to tell me what I can and can't do because some people are using it for that. So that worried me that particular. But I think generally agree, but I was like ohh that's interesting if people are then using it as a weight loss tool or a like dieting tool.

P17

Its just interesting isn't it, seeing that it could be created seeing it from a different side.

P11

Yeah, it's it's just interesting.

P04

And they're so they're still so much like there's so many people that still don't get the difference between type one and type 2 and kind of being like, “ohh, so that means you can't eat sugar” and all that stuff. And that if that's the way that it's gonna be used in mainstream industry it's gonna make that the merge together more and kind of like they'll be more of that whole “ohh no but you shouldn't be eating that cause you're diabetic” and it's like “well actually no that's not really the case”. So, I mean hopefully it's not used too much in a way that then means that they're not separate conditions because they are very, very different and are managed very differently and I'm sick of people telling me I can't eat cake.

P11

Yeah, I actually think that is a massive thing.

Interviewer

Yeah. Sorry to cut in, I’m just conscious of the time, are people alright if we go 5 minutes over or does anyone need to rush off?

P04

No.

Interviewer

Great that I that was kind of like a couple of other areas I just wanted to dive in briefly just bits that come up and that was [P17] you were saying one of the bits, this is going more back to the kind of the AI side of things, you mentioned ChatGPT and also Snapchat with that AI that brought in. I was just wondering about yeah kind of the increase of those in day-to-day life, how that's kind of shaped all your understandings of it and if that's ever kind of, kind of moved into the diabetes side and helping you understand how the technology there works?

P17

And when I first heard about AI before and this is before I even knew it was related to kind of like the algorithms of the Omnipod and stuff, I was very against it. I didn't like the idea of doing it. I think especially cause I work in a university and like with lecturers and different things as well. And there was a lot of talk about like the ChatGPT or writing assignments and things. So, it was all very negative things. And my boyfriend's absolutely obsessed with it, anything he wants a question, he uses that rather Google. I'm sick of hearing about it because it's just general AI. But then the more I kind of think any kind of thing I've using even like he was like a planning thing for like, it's a to do list in work and I know it was there's even a button on me now when it's like a summary of the information. I've put click that and AI comes up like it description. So, I think it's slowly coming into my life without me realising and so um I'm getting more kind of open to the idea um. And I think maybe from like knowing the algorithm of different things, now I'm using it and it really helpful purpose, I can see

how beneficial it can be. I think the only, I don't know that much about it but I know I think is like only data from like 2017 or something onwards that it's got access to. Um. I didn't know would it be in the same capacity isn't like a medical sense as well? I think there's a lot of questions around it but definitely more open to it.

P04

Yeah, I think I I kind of like don't really know too much about AI as a thing, but I suppose like it's probably the first thing that you always hear about like think about it with AI is things like all the big movies where it's like this evil thing and it's it's taken over the world and things like that. And I think, slowly, we're moving away from that being what AI is luckily. Um. But yeah, I don't think it will necessarily be a bad thing, it will probably just be kind of. Like it's still technology and it still can go wrong and so if it does go wrong, being able to kind of override it or um like not have to use it necessarily if it does become a problem. But I don't. Yeah, I think it's probably it is very useful in terms of diabetes management, cause it can then those algorithms can make diabetes care just a little bit less of a thing that is thought about all the time.

P11

Yeah, I found it interesting, again being a teacher, and used it a lot for like writing exercises and thinking with the kids about like how it writes. And it's it's interesting cause it's kind of only as good as the data it gets in. So, you can ask it to like write in the certain style of someone and it can do it, but often really quite clunkily and it's not quite as human yeah, as it was. Um, so I don't think it's perfect. It's interesting to see that it's accessible and people are using it, but at the moment we can still tell if that if a child has written something versus if AI has because it just doesn't sound same. So yeah, there's still a divide I think. I did, I was interested, I asked it to do like a meal plan um bearing in mind I was like type one diabetic and like I didn't like this and this and this and interestingly it came out with a whole meal plan with no carbs in. So, I think in terms of the diabetes space at least ChatGPT, it wasn't at all what I wanted. I was like “ohh, no, that's not gonna work”, because I do actually want to eat carbs and like it was still quite simplistic in the way it was thinking of like, “you know, no carbs at all is gonna make your blood sugar very stable”. And I was, you know, so I think I found that interesting cause I was hoping it would be more advanced than it actually was. Um. But I know it will get there, but it's just, yeah, I was I was kind of disappointed cause I was hoping it would be better than it was, but.

P17

I feel like with the the AI, I think cause we're all like old and we've kind of been with technology we've seen kind of the advancement of the apps. So, my brother's [brother’s age] and he loves like the idea and I don't know whether this is a gender thing as well the male female but I think as I've hit 26 and like even Tik Tok is still quite a new thing to me. Like just got my head around that. And I've always been good for tech but and like is it cause I'm get a little bit older though I'm scared of these things, like if I was younger I would ever be more open to the idea I don't know. Because my brother just doesn't care. He's like, “yeah I use it for this, I ask everything”. But it was interested about the meal plan with the carbs it is really seen that it thinks that no carbs fixes everything.

Interviewer

Yeah, definitely. I think that's really interesting point. The one, sorry, we're jumping about a bit now. The one thing that I just wanted to ask about was and this is going way, we're way back to the start was just around. Yeah, I think [P17], you were mentioning just because you liked the new tech, you kind of use it quite a lot in that initial phase and then that's maybe changing as times going on. And I guess the others, you got the Fitbit in kind of the last couple of months and how has your kind of usage of it changed over that time?

P04

I think I'm slowly kind of starting to use it more and more cause at first when I got it I was kind of like "ohh yeah I can see my bloods on it". And then that was kind of like all I used it for, whereas now I'm kind of looking at the activities for it. I've obviously got it on to start looking at my sleep profile and stuff. I think it does like it does like stress management levels and all sorts. And I like actually all of this stuff is probably gonna be really, really useful. So, I am definitely starting to see the benefits of all the things it can do, and like how that can then help me with my management of diabetes. Because I've kind of got to a point with my diabetes where I'm like I'm tech now like I'm on the best tech now that in the UK the UK has to offer probably minus Omnipod 5, but they kind of works in the same way just without the wires. So actually if my HbA1c still isn't quite as low as I want it, why is that? And what can I actually start doing to kind of bring that down a little bit more? Which I think the watch will probably, once I get used to using it for a lot more stuff, I'll be able to start seeing that and kind of like "ohh yeah I do go low every time I go on and walk" and "okay so I know I'm gonna go on a walk when my heart rate goes this high that actually my bloods are fine at that point, but then when it gets up this high, that means I'm using a lot more glucose". And so, I think I'll definitely start being able to use it more now I've got used to having a watch. And cause I definitely at first wasn't really using all the benefits that a fitness watch can offer cause I was literally just like "yay, bloods" and then it kind of just like left it at that for a little while so.

P11

Yeah, I feel like I'm probably in the same boat. Like, I feel like I'm just starting to think about how it could be helpful. I think, like I was using it quite a bit and then I got really overwhelmed and stopped and then now I kind of wanna get back into using it. But I just again, it's kind of like where to start because I know all the things it can do, but I think I probably need to be a bit disciplined and maybe sit down like once a week and look at the data. Because I just find that like day-to-day, I think I'm operating more on the watch than the app and therefore I'm not really seeing the trends and I'm I'm not really dedicating time to sit down and like look at the trends. So, I think probably that's just something I need to change about. It's not just about wearing it, but like I also need to have some time to analyse it as well to then work out what to to work on first. But yes, I feel like I'm kind of in waves, but definitely when I was away, I heard it on everyday and it was like a life saver for like seeing the bloods and then acting on it. So, yeah, I definitely. I'm I'm in and then, but I just need to work out how to deal with it on like a more reflective basis.

P17

I think we said about reflective, I started doing it just cause like most of the week I got homework. So, when I'm procrastinating and I'm scrolling on my phone. It's kind of like, instead of just going through like social media platforms, it's become like another channel to kind of look through and look through what's going on. But I think now that trend has come up or like your sugars were this range and I've started. And I think I said like my first interview, I kind of always looked at like my heart rate and things because I was saying if I know kind I'm naturally seeing if my heart rate is a lot higher, I know I'm gonna drop within half hour cause I do exercise for 45 minutes from half an hour, hour and a half. So, I know that works for me. So, it was interesting to see that and then how it will change now that I'm on less amount of insulin, will it do the same thing? So it'd be good to see, look at using that data I can see on the watch to see if that confirms my thinking.

Interviewer

Yeah, that's great to hear.